

## Unit 2

## Logarithms

| Sr. No. | Questions                                      | A                 | B                      | C                  | D                    |
|---------|------------------------------------------------|-------------------|------------------------|--------------------|----------------------|
| 1       | The standard form of $5.2 \times 10^6$ is:     | 52,000            | 520,000                | 5,200,000✓         | 52,000,000           |
| 2       | Scientific notation of 0.00034 is:             | $3.4 \times 10^3$ | $3.4 \times 10^{-4}$ ✓ | $3.4 \times 10^4$  | $3.4 \times 10^{-3}$ |
| 3       | The base of common logarithm is:               | 2                 | 10✓                    | 5                  | $e$                  |
| 4       | $\log_2 2^3 = \underline{\hspace{1cm}}$ .      | 1                 | 2                      | 5                  | 3✓                   |
| 5       | $\log 100 = \underline{\hspace{1cm}}$ .        | 2✓                | 3                      | 10                 | 1                    |
| 6       | If $\log 2 = 0.3010$ , then $\log 200$ is:     | 1.3010            | 0.6010                 | 2.3010✓            | 2.6010               |
| 7       | $\log(0) = \underline{\hspace{1cm}}$ .         | positive          | negative               | zero               | undefined✓           |
| 8       | $\log 10,000 = \underline{\hspace{1cm}}$ .     | 2                 | 3                      | 4✓                 | 5                    |
| 9       | $\log 5 + \log 3 = \underline{\hspace{1cm}}$ . | $\log 0$          | $\log 2$               | $\log \frac{5}{3}$ | $\log 15$ ✓          |
| 10      | $3^4 = 81$ in logarithmic form is:             | $\log_3 4 = 81$   | $\log_4 3 = 81$        | $\log_3 81 = 4$ ✓  | $\log_4 81 = 3$      |

### Solution of MCQs

|    |                                                                                                           |
|----|-----------------------------------------------------------------------------------------------------------|
| 1  | $5.2 \times 10^6 = 5,200,000$                                                                             |
| 2  | $0.00034 = 3.4 \times 10^{-4}$                                                                            |
| 3  | Common log base = 10                                                                                      |
| 4  | $\log_2 2^3 = ?$<br>$\log_2 2^3 = 3 \log_2 2$<br>$= 3(1)$<br>$= 3$                                        |
| 5  | $\log 100 = ?$<br>$\log 100 = \log 10^2$<br>$= 2 \log 10$<br>$= 2(1)$<br>$= 2$                            |
| 6  | $\log 200 = ?$<br>$\log 200 = \log 2 \times 100$<br>$= \log 2 + \log 100$<br>$= 0.3010 + 2$<br>$= 2.3010$ |
| 7  | $\log(0)$ is <b>undefined</b>                                                                             |
| 8  | $\log 10,000 = ?$<br>$\log 10,000 = \log 10^4$<br>$= 4 \log 10$<br>$= 4(1)$<br>$= 4$                      |
| 9  | $\log 5 + \log 3 = \log 15$                                                                               |
| 10 | $3^4 = 81 \Rightarrow \log_3 81 = 4$                                                                      |